JESAL DEEPAKBHAI SHAH

jesalshah1510@gmail.com Jesal Shah shahjesal15

EDUCATION

M.Sc. in Engineering Cybernetics
Universität Stuttgart — Sep 2026 – Present

B.E. in Electronics and Communication — CGPA: 2.1/5.0
Gujarat Technological University — Jun 2024

Thesis: *Computed torque control based trajectory tracking for legged locomotion*

PROFESSIONAL EXPERIENCE

Associate Robotics Engineer — Sept 2024 – April 2026
Strider Robotics — Bengaluru, IN

- **Migrated** a quadruped's legacy ROS1 state estimator to ROS2 to support the robot's updated software stack, **validating** the Kalman Filter's output against motion-capture ground truth to confirm estimation accuracy.
- **Developed** an internal ROS2 SDK and **standardized** robot software configurations, **eliminating** manual setup steps and significantly reducing bring-up time for control and navigation experiments.
- **Designed** and **implemented** push- and fall-recovery behaviors for a quadrupedal robot, **enabling** it to withstand slips and external pushes and to self-right from a fully inverted (upside-down) position.
- **Estimated** motor and gearbox parameters using CMA-ES and **updated** Isaac Lab's actuator model to match the real quadruped, narrowing the sim-to-real gap for RL policy training.
- **Trained** an actuator-net on real hardware data to model motor torque nonlinearities; together with the calibrated simulation, **reduced** thermal load by 25% and **eliminated** overheating-related hardware failures during slope walking.
- **Migrated** the quadruped simulation from the end-of-life Gazebo Classic to Gazebo Harmonic and **ported** internal custom plugins to the gz-sim framework, **restoring** full simulation support.

Robotics Intern — Jan 2024 – Aug 2024
Strider Robotics — Bengaluru, IN

- **Integrated** the Unitree SDK for the GO1 quadruped, **enabling** rapid hardware deployment and testing.
- **Implemented** a model-based computed torque controller for quadruped leg trajectory tracking, **reducing** mean tracking error by 10% relative to a PD controller tracking baseline.
- **Developed** actuator-specific ros2_control hardware interfaces and controllers to test multiple QDD actuators on a common setup, **benchmarking** speed, torque, and efficiency to inform actuator selection.

LEADERSHIP AND ACTIVITIES

Software Lead — Aug 2022 – Nov 2023
GTU Robotics Club (GRC 2023) — Ahmedabad, IN

- **Led** a software team to build a modular stack with omnidirectional navigation and dead-reckoning localization, **winning** DD Robocon 2023 (1st of 63) and **representing** India at ABU Robocon 2023 (6th of 13).

OPEN SOURCE CONTRIBUTIONS

RViz Dynamic Reconfigure — Maintainer

- **Developed** an **RViz2** plugin for live GUI-based reconfiguration and logging of **ROS2 node** parameters, **streamlining** tuning and debugging.

SKILLS

Programming Languages	:	C++, Python, Bash
Software/Tools	:	ROS, ROS 2, CMake, Git, Gazebo, MuJoCo, Nvidia Isaac Lab, Moveit2, RViz2, Nav2